

File / vector I/O

Regression stimulus often lives outside the testbench source

Stimulus, expect, and apply loop

- The usual shape is three arrays: stimulus, expected, and optionally actual results
- Dollar-readmemh loads hexadecimal words line by line
- A for loop walks from zero to N minus one
- The same loop handles four vectors or four thousand

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

File / vector I/O

Treat a hex file as TB stimulus: \$readmemh into an array, apply vectors to a tiny DUT, compare against an expect file. Starter: four and2 cases loaded from stim.hex / exp.hex.

Starter example: four and2 vectors from stim.hex / exp.hex — only 03 expects y=1.

Load starter example

Challenges 0 / 22 cleared

Quiz: \$readmemh: \$readmemh is typically used to...

☐ load hex words from a file into a memory/array

☐ compile SystemC only

☐ replace \$finish

☐ draw GTKWave cursors

Show hint

Check

Next

Idle

1

2

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Core ideas

\$readmemh

Load hex words from a file into a memory / vector array.

@addr

Optional address marker — next word starts at that index.

Apply loop

Drive DUT from stim[i], compare to exp[i].

Separate data

Vectors live in files so TB code stays small and reusable.

Browser lab starter

learn_sv_tb — file vector io

Real SystemVerilog practice

- In the real SystemVerilog track, open this module's examples prompts
- Restate the apply-loop idea in one sentence , load hex files, drive, compare
- Sketch the pattern below on paper and label what each array holds
- Optional stretch
- ```systemverilog // Load vectors from hex files at time zero reg [7:0] stim [0:DEPTH-1]`

Pitfalls to watch

- Do not mix up `dollar-readmemh` and `dollar-readmemb`
- Do not forget `at-address` when your files have gaps or comments
- Do not compare before inputs have settled
- And remember

Your turn

- Complete the checklist for at least one track , preferably both
- In the browser, load the starter and-two set, run Apply all
- On paper, write two hex lines for stim and two for exp that test an or-two
- When you are ready